

A Visual Marker based Image Annotation Tool with Projective Transformation

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Abstract—Although template matching has been studied for decades, difficult problems still exist, such as template matching with projective transformation. Projective template matching is challenging not only because of its own hardness, but also owing to the high cost in creating ground truths. Focusing on the problem in creating ground truths, in this paper, we propose a novel method to automatically create ground truth by augmented reality (AR) technology. Specifically, when creating dataset, around each target image we attach AR markers. Then we shoot videos containing target image with the markers from different angles and backgrounds. Finally we locate the target images by detecting the markers. We evaluate our method by deviation, intersection over union (IoU), and necessary correction rate, indicating that our method is available.

Index Terms—ground truth creation, template matching, AR marker, automatic annotation

I. INTRODUCTION

A. Projective Template Matching

Template matching is to find small parts of a digital image that is similar to a template image. It is utilized by many categories of industries in manufacturing, such as quality control. In image processing, template matching can be also regarded as an approach to detect edges in images.

From simple to complex, template matching can be classified into 4 kinds of problems, including Euclidean matching, similarity matching, affine matching, and projective matching, respectively corresponding to 3, 4, 6, 8 degrees of freedom (DoFs) [1]. The Euclidean template matching only controls translation and rotation, while similarity template matching adds the control of scale. These two kinds are simple and fast owing to their low dimension of DoFs. However, they do not satisfy some cases (e.g., images taken by wide-angle lens exist affine transformation and projective transformation). In order to accurately match the template images, DoF of affine matching grows to 6 by separating scale to vertical and horizontal scale, and increasing the angle of affine transformation. For projective template matching, there is an absolutely different definition of DoFs. Certainly, the 8 DoFs should include double directions of translation and scale. It also includes 3D based rotations. The remained DoF is the distance between view point and the target image.

To create the ground truths of the Euclidean matching and similarity matching, we only need to draw a rectangle and adjust its rotation. For affine matching, it requires one more angle to adjust, and is complex. Further, in case of projective

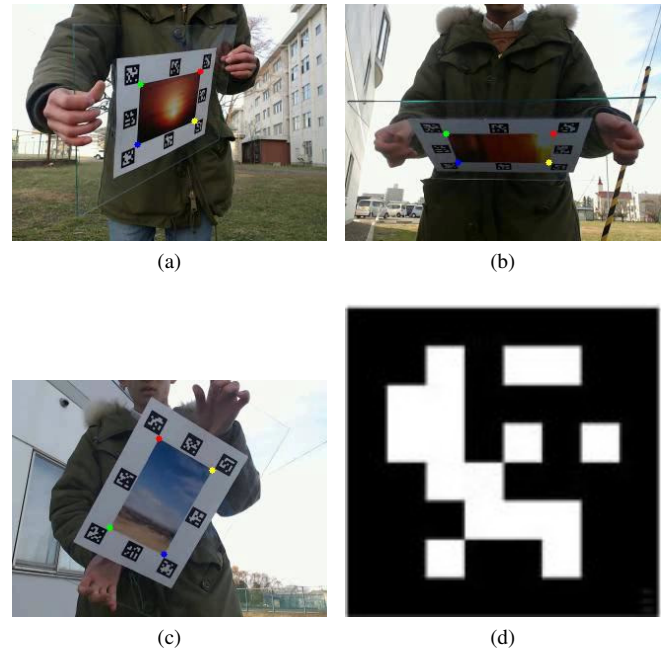


Fig. 1: (a)~(c): Ground truths created by AR marker with raw, yaw and pitch rotations, especially, (b) is taken in dark environment. The points are target image location detected by our method. (d): Example of AR marker.

matching, it is more difficult to create ground truths according to DoFs. Therefore, usually people mark 4 points directly when creating ground truths. It is a complicated operation that we have to click four times for each target image, still less the huge numbers of dataset images. Therefore, an automatic image annotation tool is necessary for template matching.

B. AR Marker

AR is increasingly popular in researches, games, and applications. AR is a live view of a real-world environment, but some of its elements are augmented by information that generated by computer. AR marker is an application of AR technology, which is a 2D symbol. A traditional AR marker looks like a quick response code (QR code), the detection method of which is much different from it. AR marker allows cameras to detect the location and rotation relative to a surface.

Nowadays, there are many new categories of AR marker, such as logo marker and object marker. However, the traditional QR-code-like AR markers are still the most promising markers to determine location and rotation.

C. Contributions

In this paper, we propose to create ground truths automatically by AR markers for template matching. Specifically, we have the following contributions. 1) We propose an automated ground truth creation system. 2) We create a dataset for projective template matching with different environments and target images. 3) We explore multiple effects on AR marker detection and optimize it.

II. RELATED WORK

To the best of our knowledge, there is not any research that is the same to this paper. However, researches on template matching and AR marker detection concern our method.

A. Template Matching

The Euclidean and similarity template matching has already been applied because of the fast speed and high accuracy, e.g., Akashi et al. [2] apply similarity template matching to eye detection, Lin et al. [3] propose to detect human by template matching. Our method can support these researches in creating new datasets.

Multiple researches attempt to overcome the effect of environment. Grabl et al. [4] try to overcome the influence of illumination. Valli et al. [5] propose to reduce noise for template matching by asymmetric correlation. Itamar et al. [6] propose a template matching method that works not only in different environments, but also with rotation. Our method can help the above researches much. On one hand, even the state-of-the-art of these researches do not reach 100% accuracy, they still need new datasets for evaluation so as to be improved. On the other hand, our method supports the change of environment and 3D rotation.

The most popular researches on template matching now are affine and projective template matching. Simon et al. [7] propose a affine template matching method named FAsT-Match, which minimizes the Sum-of-Absolute-Differences (SAD) error measure. Zhang et al. [8] improve the FAsT-Match by a binary Galois field. Zhang et al. [1] propose a matching method that involves projective transformation. Our ground truth creation can also help these categories of researches, because using our method can create especial dataset for evaluating in various conditions.

B. AR Marker Detection

We consider to rely on AR marker detection because of its high success rate. Hirokazu et al. [9] propose a marker detection and tracking method. Martin [10] propose a marker detection methods for AR applications, with improving the accuracy. Koch et al. [11] evaluate the performance of several kinds of natural markers detection. All of the researches on AR marker detection improve the accuracy, benefiting our method.

III. GROUND TRUTH CREATION

A dataset for projective template matching should contain 3D rotation and complex environments. Furthermore, as is stated in section I-A that projective template matching involves the distance between a camera (view point) and the target image, it also should be considered to create a dataset.

We attach 8 AR markers around a target image. The markers form a rectangle, which is a little larger than target image, because adjoining can effect template matching and marker detection. The target image with AR markers are printed on a piece of A4 paper, which is attached to a transparent plastic board that is much larger than A4 size, as is shown in Fig. 2. When we take videos of the target image, the board is rotated manually. Therefore, the transparent board help to avoid the effect of human hands.

We collect 18 target images from imageNet [12]. For each image, after conducting the above operations, we take videos in different environments and the distances with 3D rotations. Specifically, each target image involves 4 kinds of rotations, including main raw, yaw, pitch rotations, and mixed rotation. Examples of raw, yaw, and pitch rotations are shown in Fig. 1(a)~(c).

The AR markers help to locate the target image. The accuracy of AR marker detection is relatively promising, and much more efficient than manual annotation. Thus we promote accuracy of AR marker detection and determine the ground truth according to it. However, a few frames fail to locate the target image by AR markers. To overcome these a few frames, we check the videos manually and adjust the results, which cost much less human resource and time than annotating all the frames.

IV. AR MARKER DETECTION

In this section, we introduce the basic AR marker detection method and discuss the effects of the detection.



Fig. 2: Transparent board attached with the paper, on which the target image and AR markers are printed.

Number of markers	Deviation (pixel)
1	7.05
2	2.20
3	1.24
4	0.97
5	0.91
6	0.88
7	0.86
8	0.85

TABLE I: Effect on AR marker detection of number of markers

A. Basic Method

Mainly our method is based on AR marker detection, which is the most important portion. On basis of section II, AR marker detection has been well studied and applied. We select [9] as our basic method. Besides, we process each frame by Canny edge detector [13] so as to promote the accuracy.

B. Number of Markers

In this section, we discuss the effect from the number of markers. We locate the markers at the vertices and at the middle of the longer edges, totally 8 markers. In case of multiple markers, we compute the average coordinate. We evaluate the effect of the number of markers by compare the detected target image location with ground truth. The comparison is based on deviation (pixel). In our ground truth creation system, the number of conditions is

$$\sum_{i=1}^8 C_8^i = 256. \quad (1)$$

The evaluation of the number of markers is listed in Tab. I. The deviation decrease with the increasing of the number of markers. For a certain frame, if the number of markers is less than 3, there is a more than 1 pixel deviation in locating target image. If the number of markers is more than 5, the decrease rate becomes lower, especially, deviation between 7 markers and 8 markers exist only 0.01 pixel. In conclusion, more AR markers can lead to higher accuracy, but when the number of markers grows to 5, it becomes available, when the number of markers grows to more than 7, it is relatively promising.

C. Available Range of Markers

Although AR marker detection has high accuracy, it is limited by rotation and distance within a range. The rotation limits mean it cannot be detected if the angle from view point (camera) is too large. That the distance between view point and markers is too large results in that the markers are too small to be detected.

We conduct several comparison experiments to explore the available range. We rotate by pitch and yaw the transparent board from -60° to 60° , at the interval of 15° . We also

modify the distance from 50 cm to 100 cm, at the interval of 10 cm. Besides, because we exploit Canny edge detector for preprocessing, the parameter of Canny is also important in marker detection. We define B denotes the block size in Canny, and C denotes threshold in Canny.

The evaluation is described in Fig. 3. We can conclude that a larger distance or rotation can result in lower accuracy. In the situation that $B \in [17, 29]$ and $C \in [4, 16]$, the detection is accurate.

We analyze in section IV-B that more than 5 markers are available to locate the target image, and more than 7 markers are promising. According to Fig. 3, we describe the warning range and promising range by diagrams in Fig. 4, where the warning range represent the case that more than 5 markers can be utilized, and the promising range represent the case that more than 7 markers can be utilized.

D. Reversal Detection

In AR marker detection, one of the common errors is reversal detection. Detecting AR marker is to detect the location and 3D direction. Reversal detection means the reverse detection of 3D direction, as is shown in Fig. 5. In general, reversal detection happens when an AR marker rotates. However, it also happens in static cases. This error decrease the accuracy of target image detection.

We consider reversal detection in static cases is related to the distance between camera and markers and the parameter of edge detector (B and C).

1) *Relevance to Shooting Distance*: Since the videos taken in front view can be detected with a high success rate, containing few reversal detection, we conduct the comparison experiment in the condition that pitch rotation is 30° to explore the relevance between reversal detection and distance. Figure 6(a)~(d) display the comparison from 70 cm to 100 cm shooting distances. Each of the diagram is on basis of multiple static videos in constant distance and rotation, and in different environments. For drawing the diagrams, we modify B and C for the videos, with take notes of the reversal detection. The number of reversal detections is normalized and described by color.

We can conclude from Fig. 6(a)~(d) that with the growing up of the distance, the diagrams become increasingly bright. It indicates that in static environment, high shooting distances is one of the reasons that result in reversal detections.

2) *Relevance to Edge Detector*: We conduct the same category of experiment to the above for exploring the relevance between edge detection parameters and reversal detection. However, change of reversal detection with modifying B and C need to be supported in plenty of situations that involves a variety of rotations and shooting distances. Thus we take videos from the rotation range and shooting distance range that are the same to section IV-C. In this paper, we only display representative diagrams in Fig. 6(e)~(h).

We can conclude that in most diagrams, places that are more close to the lower-right corner, turn out to be darker. It

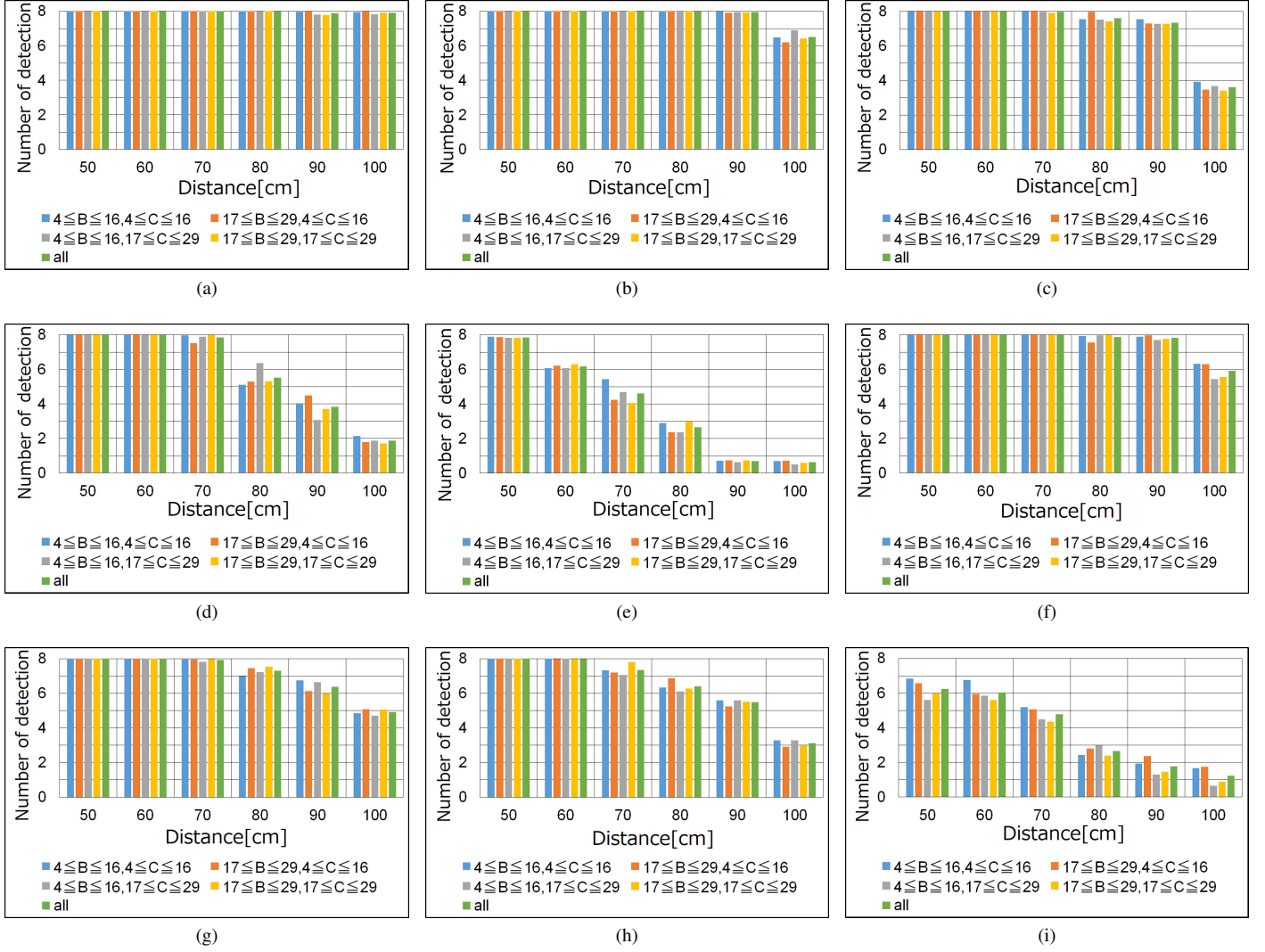


Fig. 3: Evaluation of effect on AR marker detection of rotation, distance and edge detector parameters.

indicates that if we modify B higher and C lower, the reversal detection decreases.

V. EVALUATION

A. Dataset

The evaluation of our method is based on such dataset that created by us for projective template matching, in which the videos are shot in several kinds of indoor and outdoor environments. Specifically, indoor videos are shot in lobby, classrooms and computer rooms, without any additional illuminants or backdrops. Outdoor videos are taken in complex backgrounds, such as in shine and cloudiness. However, the outdoor videos are not taken in night and rain. The dataset includes multiple template image. For a certain pair of template image and environment, we shoot four categories of videos-raw rotation mainly, yaw rotation mainly, pitch rotation mainly and mixed rotations. Besides, each video is checked and corrected manually.

B. Intersection over Union (IoU)

This evaluation approach is also known as Jaccard index. For a pair of sets P and Q , the IoU of them can be calculated by

$$J(P, Q) = \frac{P \cap Q}{P \cup Q}. \quad (2)$$

Thus IoU is a floating number and $J \in [0, 1]$. In our case, P and Q are respectively results detected by our method and ground truth marked by hands.

The results are shown in Tab. II. There exist not any result of which IoU is lower than 0.90. Results converge on $[0.97, 0.99]$, with the proportion of 88.61%. It represents that the location by AR markers are almost the same to ground truth.

C. Deviation

In section IV-B, we discuss the number of AR markers by deviation, as well. In this section, we evaluate our method

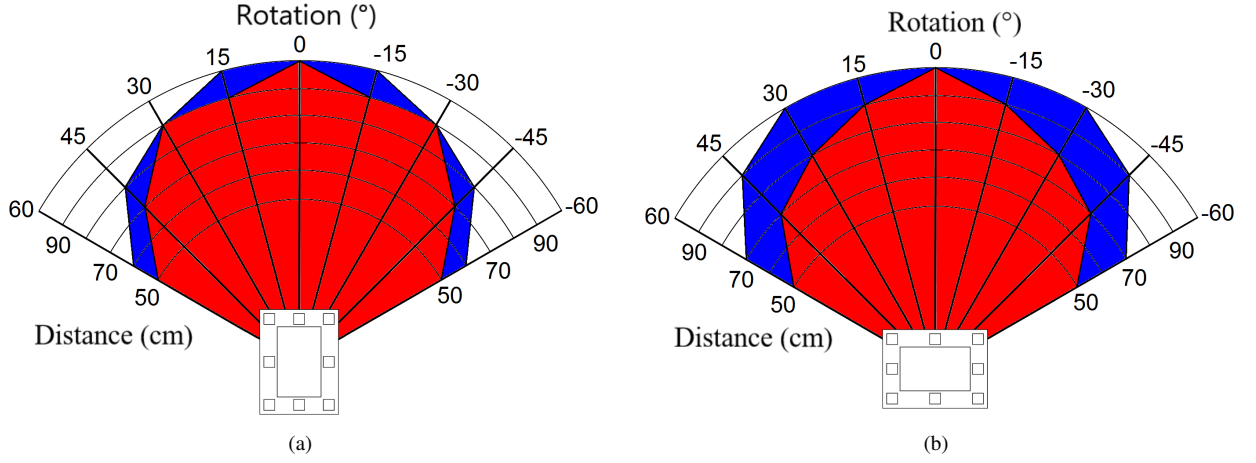


Fig. 4: Available video shooting range. The range illustrated by blue is warning range. The range illustrated by red is promising range.

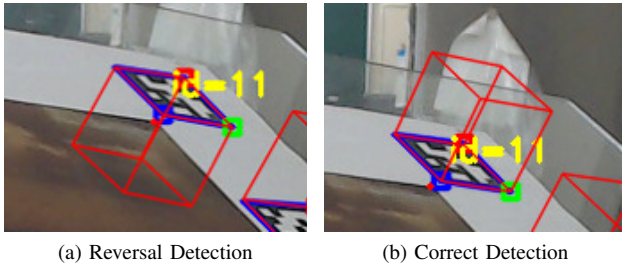


Fig. 5: Introduction of reversal detection. Result in (a) is reversal detection, while result in (b) is correct detection.

by the deviation of each vertex. Ground truth of projective template matching is a quadrilateral, owning 4 vertices. We display the average deviation (pixel) of the 4 vertices by x and y axis direction in Tab. III. The vertex number is determined according to the vertex position of target image.

By Tab. III, we can summarize that deviations of vertices are not far from each other. Besides, average deviation of all the vertices is 0.93 pixel, which is rather low. Results located by our method is extraordinarily close to ground truths.

IoU	Proportion
[0.00, 0.90)	0.00%
[0.90, 0.96)	1.94%
[0.96, 0.97)	5.00%
[0.97, 0.98)	28.33%
[0.98, 0.99)	60.28%
[0.99, 1.00]	4.44%

TABLE II: Evaluation by IoU.

D. Necessary Correction Rate

We state in section III that our method remains a few frames that need to be adjusted manually. We compute the appearance rate of such frames, named necessary correction rate. Indoor environment is much different from and easier than outdoor environment, so we list the two conditions respectively. The results are shown in Tab. IV. As we expect, indoor necessary correction rate is lower than outdoors. To observe the results more directly, we also add up the sum number of frames and the number of frames that need to be corrected. The sum frame number is 58322, while the number of frames need to be corrected is 394.

VI. CONCLUSION AND FUTURE WORK

We propose automated ground truth creation method for template matching by AR markers. We attach AR markers around the template image and shoot videos of the image in different backgrounds. We detect AR markers to locate the template image. We explore several possible effects on AR marker detection through experiments. Further, we evaluate our method by IoU, deviation and necessary correction rate, indicating our method has a high accuracy and can save human resource.

For future work, we plan to focus on the approach to delete the probable error AR marker detection. So far we calculate the

Vertex	Average deviation (pixel)	
	x	y
1	0.68	1.23
2	1.33	1.08
3	0.61	0.67
4	1.16	0.71

TABLE III: Evaluation by deviation.

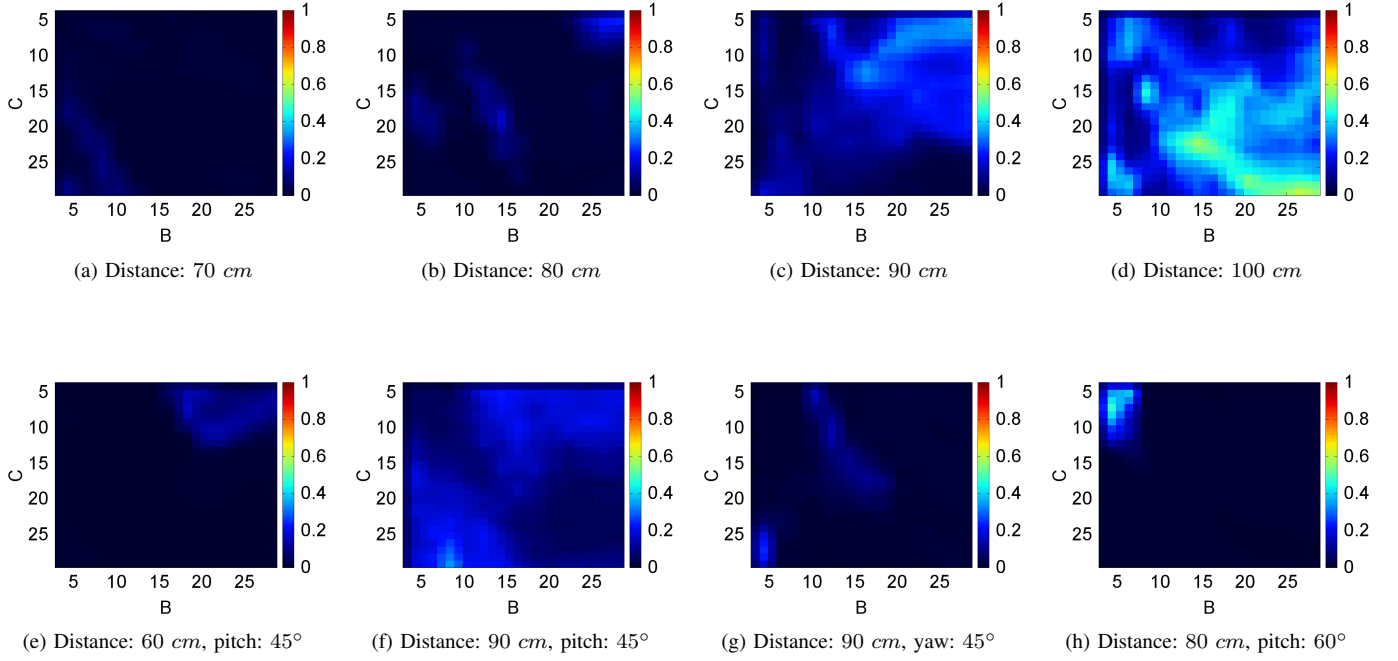


Fig. 6: Effects on reversal detection. In each diagram, x axis denotes B , y axis denotes C , and the number of reversal detections is described by color (z axis). The number of reversal detections is normalized within $[0, 1]$. More bright places means there are more reversal detections. Diagram (a)~(d) are to explore the relevance to shooting distance, all of which are with a 30° rotation by pitch, while (e)~(h) are to explore the relevance to B and C .

Environment	Necessary correction rate
Indoor	0.47%
Outdoor	0.94%
Average	0.63%

TABLE IV: Evaluation by necessary correction rate.

mean location of all the markers, instead of correctly detected markers, which may decrease accuracy. Therefore, we plan to propose an approach to judge the correction of AR marker detection.

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